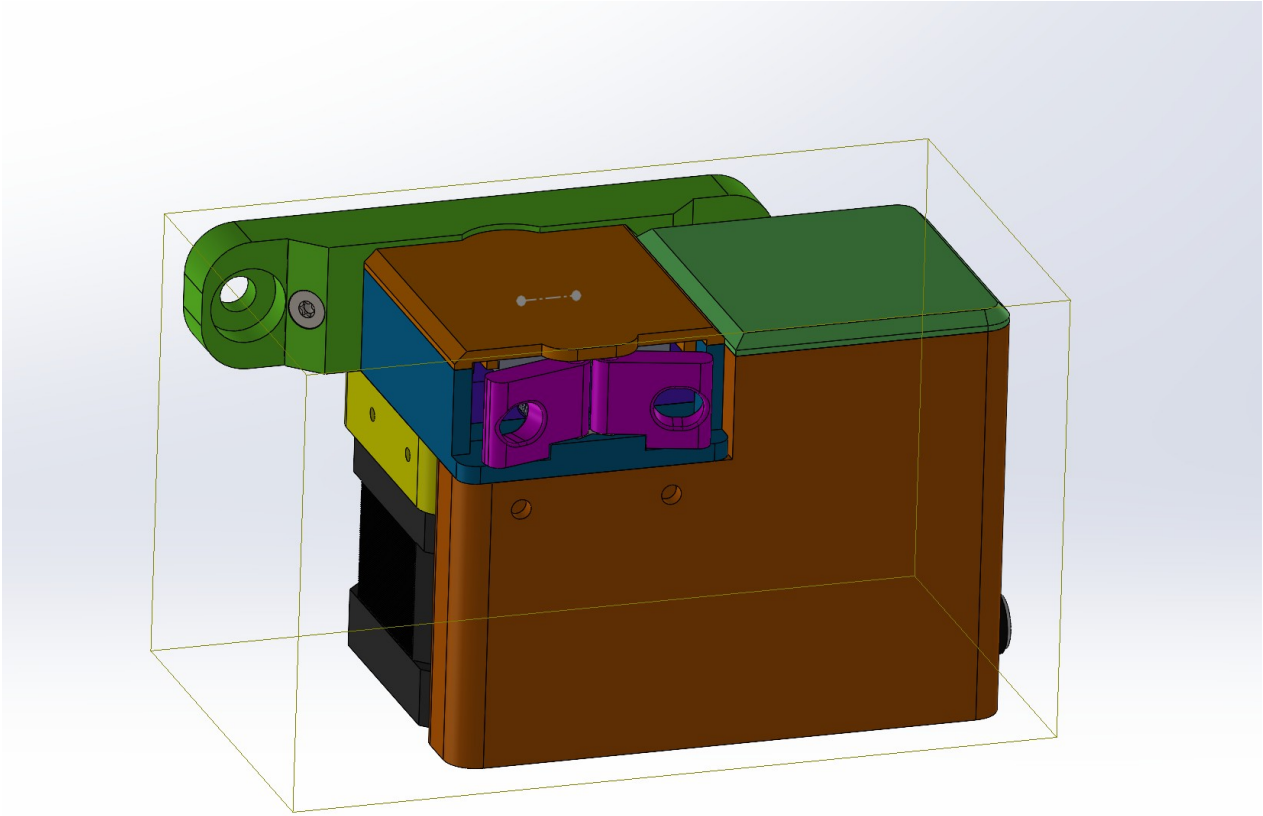
BUILD MANUAL

Assembly Guide

Step-by-step instructions based on the original author’s text, illustrations and parts list.



Assembly overview (original drawing)

11 assembly steps | 24 BOM items | Original drawings retained

PREPARATION

Parts & hardware

Item numbers are the references used in the exploded assembly drawing and throughout this guide.

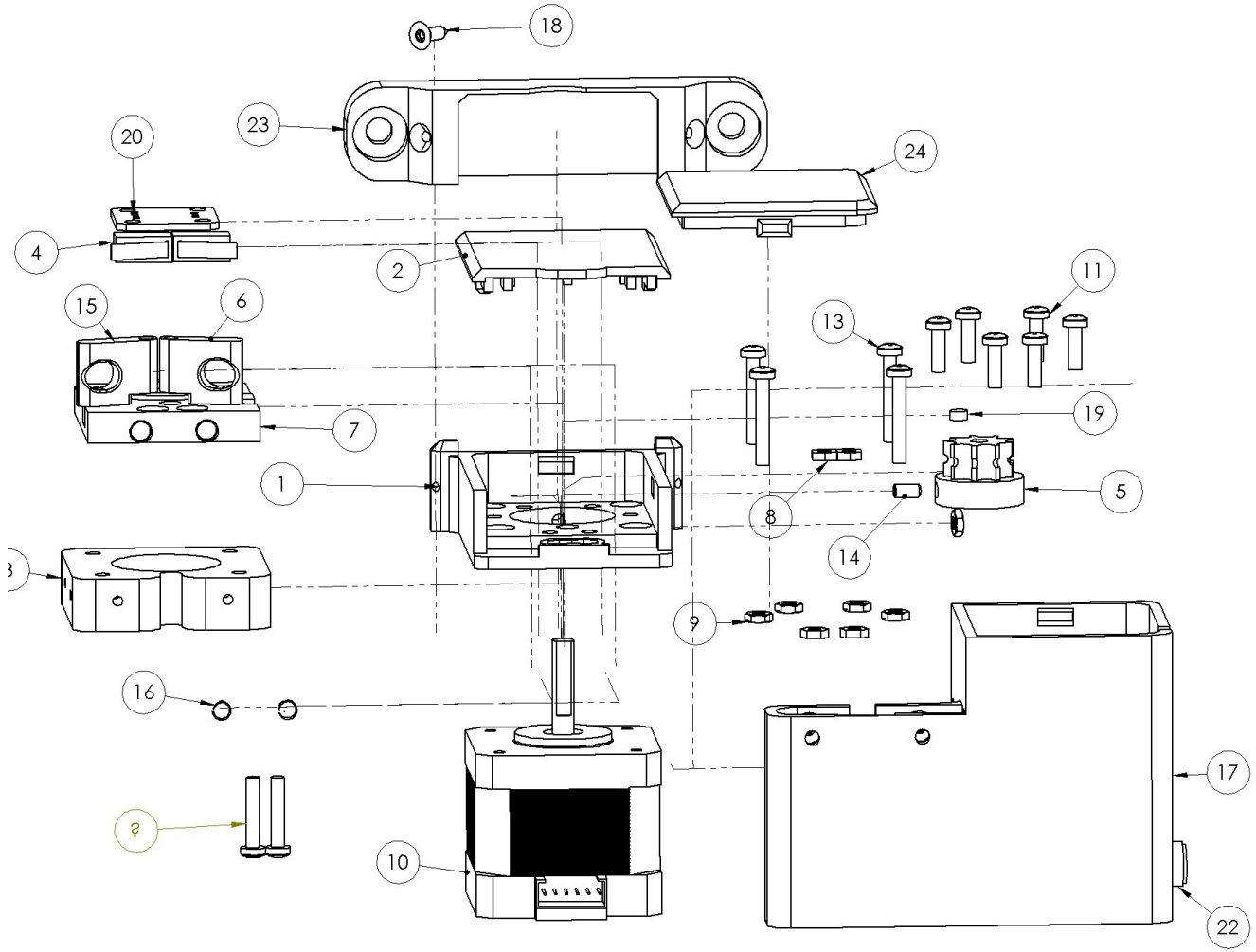
ITEM	PART / HARDWARE	QTY	ITEM	PART / HARDWARE	QTY
1	Body	1	13	M3 × 20	4
2	CoverGearbox	1	14	Grub screw M3 × 6	1
3	NemaExtender	1	15	MirrorGateBall	1
4	End Switch	2	16	Spring	2
5	gear9	1	17	electronicBox	1
6	Gate	1	18	Self-tapping screw 2.5 × 5	4
7	Separator	1	19	Magnet	1
8	M3 Nut	4	20	AS5600	1
9	M3 Nut	5	21	ESP32	1
10	Nema 17HS4401-S	1	22	DC Jack	1
11	M3 × 10	6	23	WallMount	1
12	M3 × 16	2	24	electronicBoxCover	1

NOTE Items 8 and 9 are both named “M3 Nut” in the supplied BOM. Their item numbers are kept distinct here; the source does not specify their differences.

REFERENCE DRAWING

Exploded assembly view

All item numbers refer to the Bill of Materials on the preceding page.



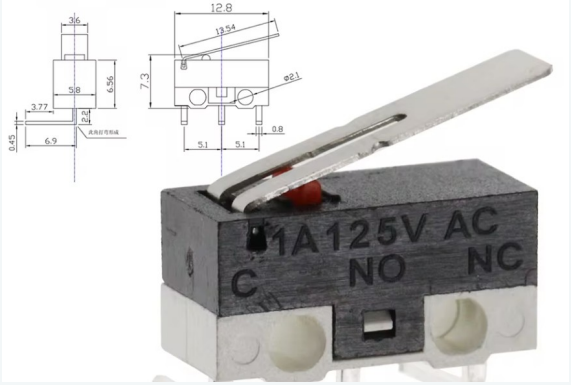
Original exploded drawing — part numbers preserved

IMPORTANT BEFORE YOU START

Check the component versions

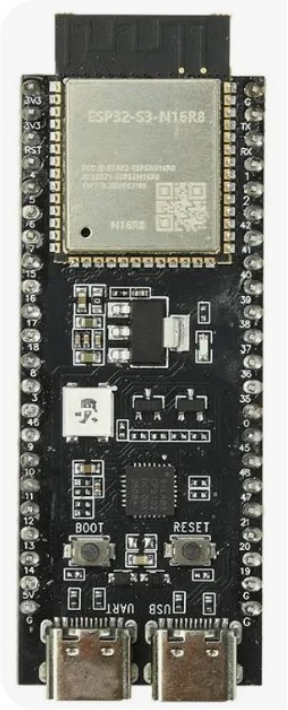
The enclosure was designed around the specific component models illustrated below. Check the physical form factor before assembly.

END SWITCH (4)



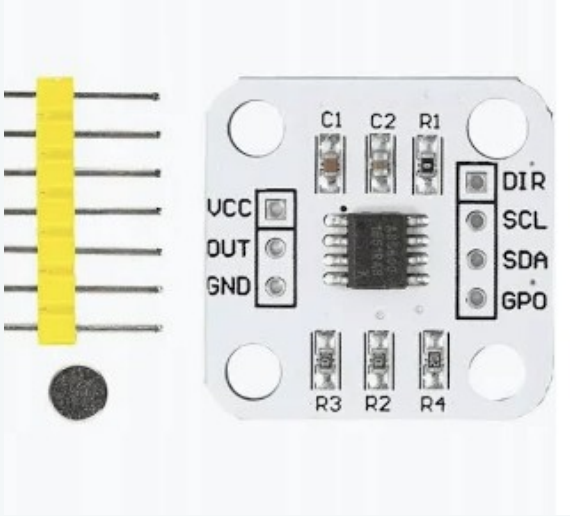
The switch shape and terminal arrangement should match the reference.

ESP32 BOARD (21)



The USB connector and header-pin orientation matter in Step 11.

AS5600 (20)



The sensor board must fit the CoverGearbox mounting area.

STEPS 01–02

Prepare the body and mount the motor

01 PRESS-FIT THE NUTS

Press-fit the M3 nuts into the matching recesses in Body (1).

02 INSTALL THE MOTOR

Slide NemaExtender (3) onto the Nema 17HS4401-S motor (10).

Place Body (1) over NemaExtender (3). Insert the washers and four M3 × 20 screws (13), then tighten the screws.

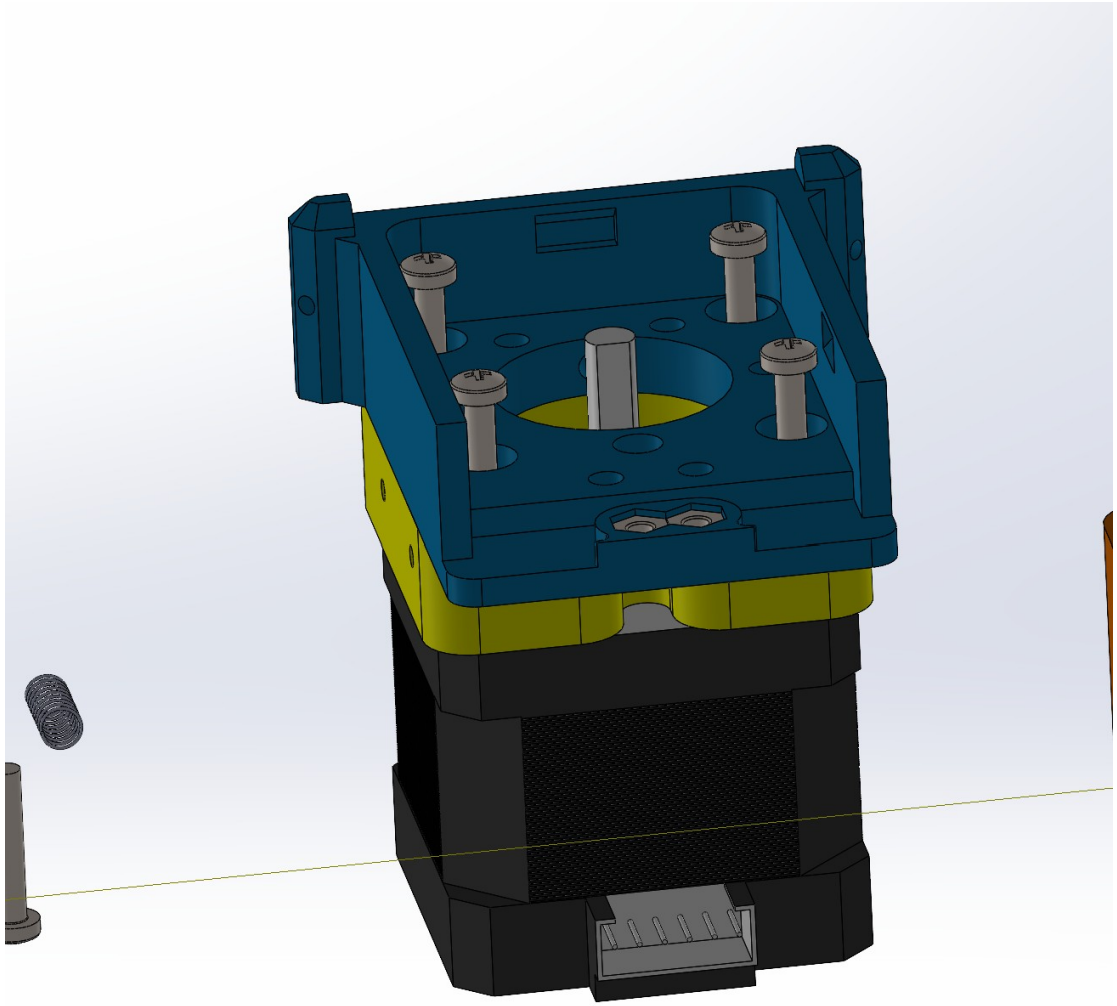


Figure 2 — motor, extender, Body and four screws

STEP 03

Install the two M3 × 16 screws

Install the two M3 × 16 screws (12) at the positions shown in the drawing.

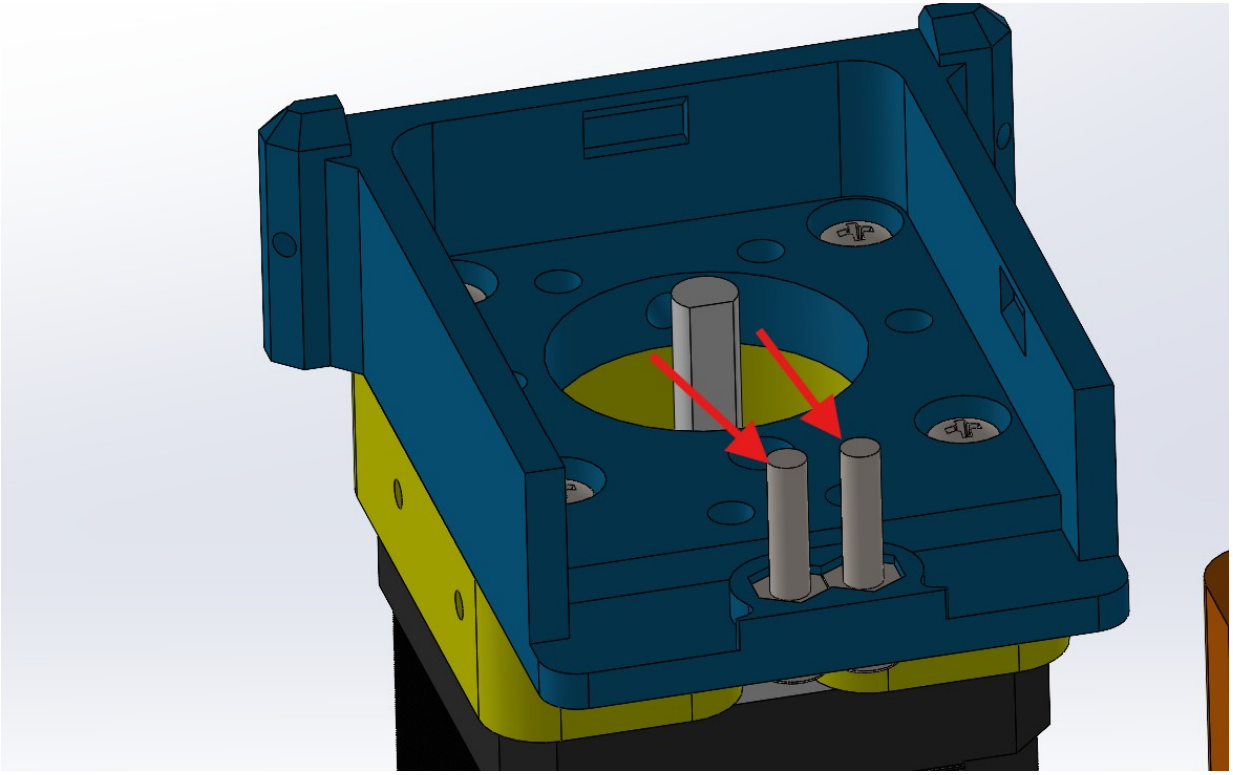


Figure 3 — screw positions

STEP 04 • PART 1 OF 2

Attach electronicBox

Position electronicBox (17) against Body (1).

Start one or two self-tapping screws (18). One screw is sufficient to hold the enclosure in place initially.

Use a screwdriver that fits through the access opening in Body (1) to support and tighten the screws, as shown.

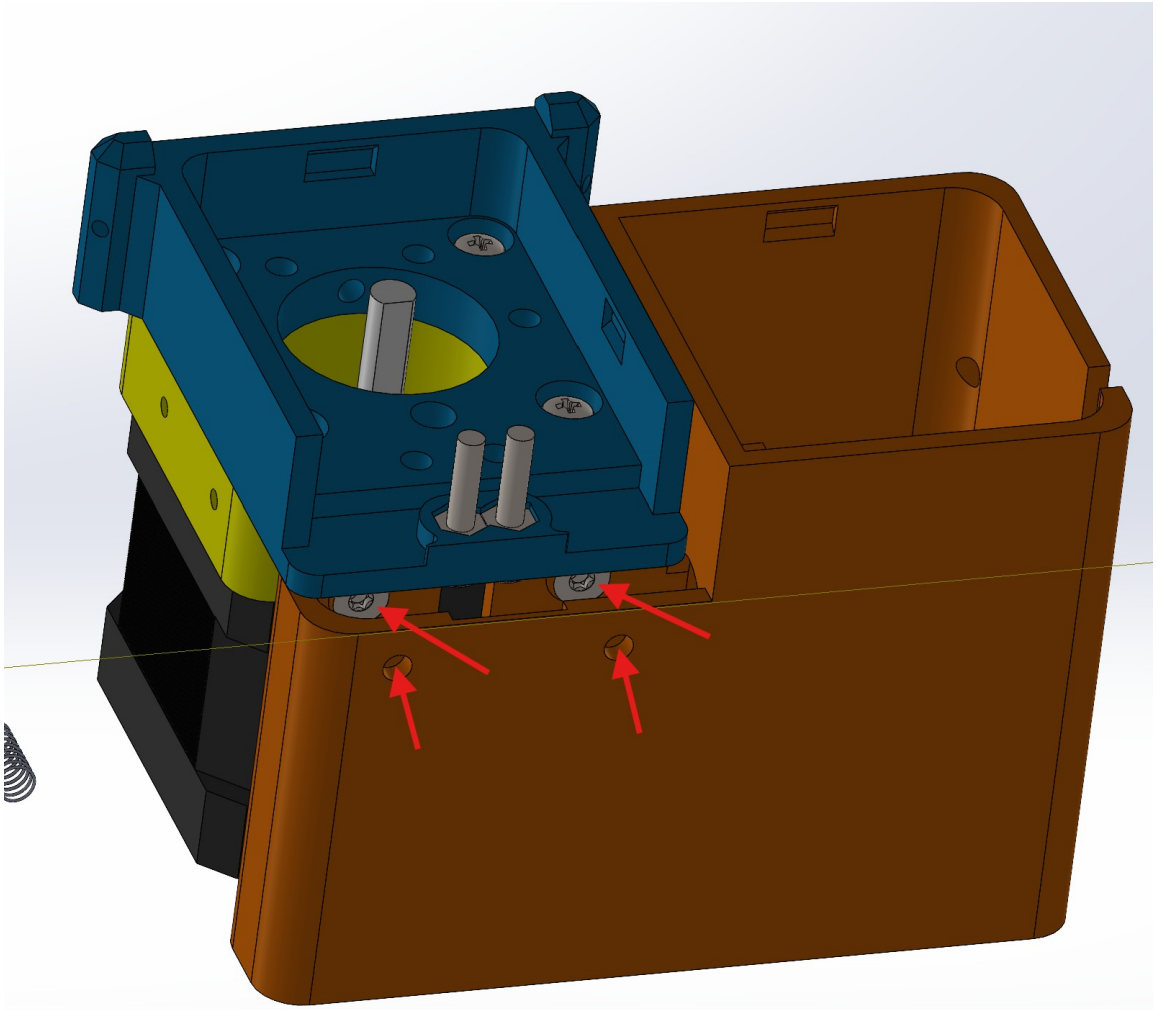


Figure 4 — fasten through the access holes

STEP 04 • PART 2 OF 2

Secure the opposite side

Install another self-tapping screw (18) from the other side of electronicBox (17), as shown.

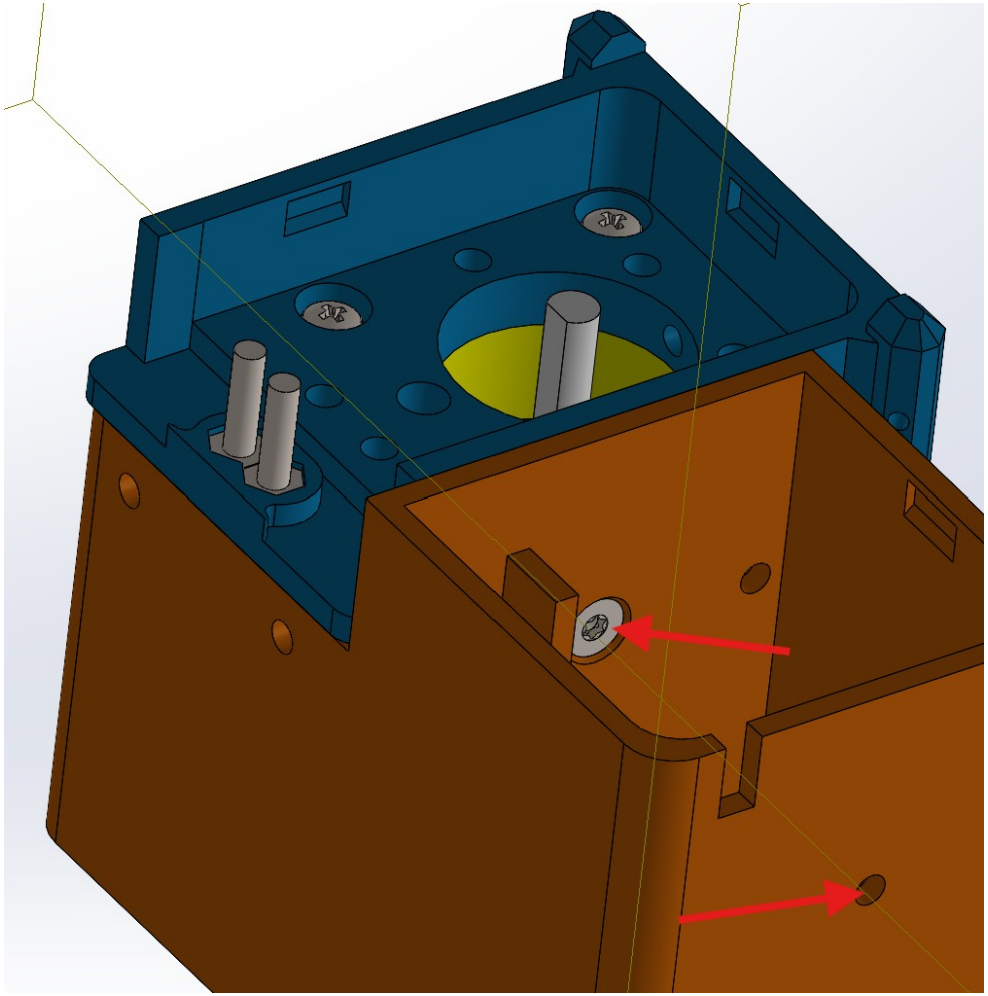


Figure 4a — screw on the opposite side

STEP 05 • PART 1 OF 3

Prepare gear9

Insert an M3 nut into gear9 (5).

Thread the M3 × 6 grub screw (14) into the nut, as shown.

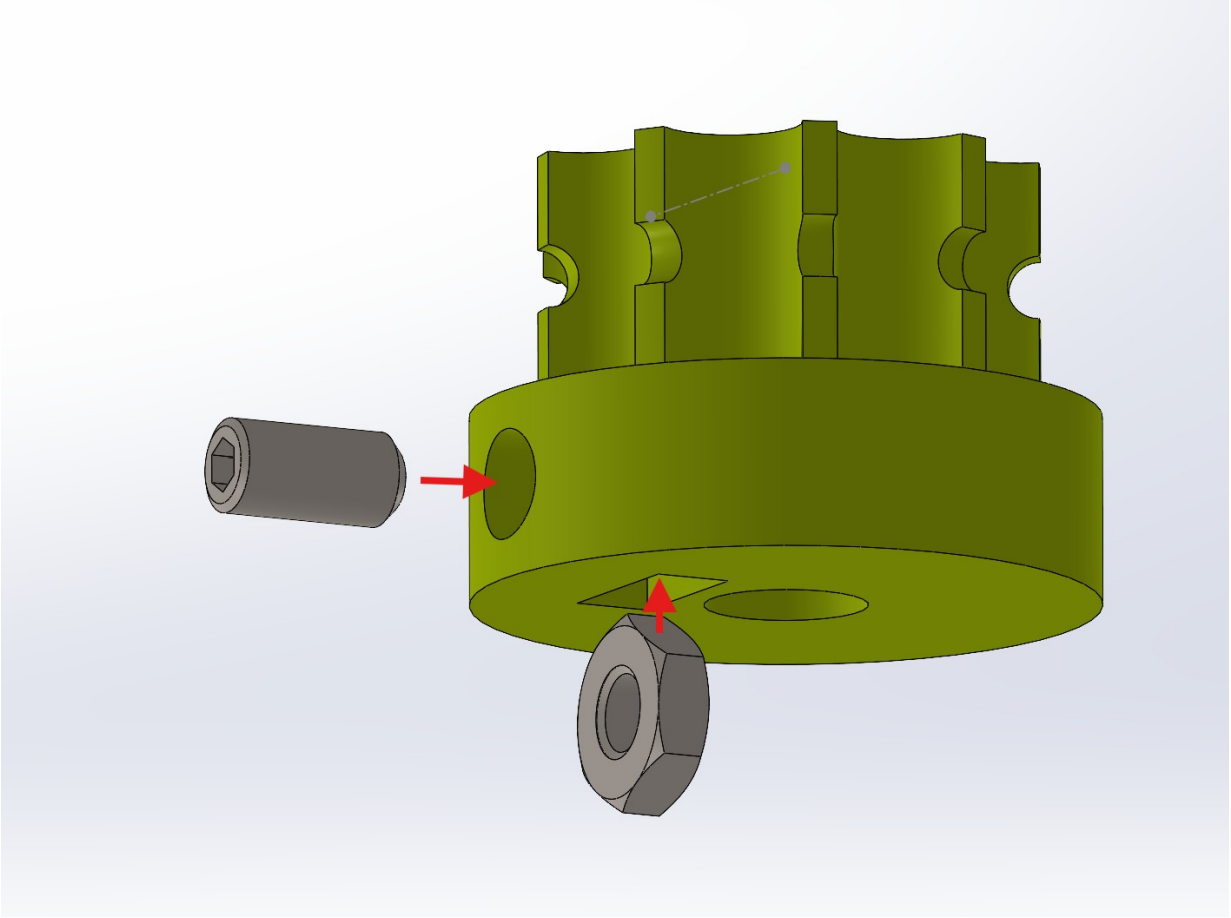


Figure 5 — nut and grub screw in gear9

STEP 05 • PART 2 OF 3

Position the gear on the motor shaft

Slide gear9 (5) onto the motor shaft.

Lower the gear until its lower edge sits approximately 0.5 mm inside Body (1), as illustrated.

POSITION

Keep the approximate 0.5 mm offset indicated in the original instructions.

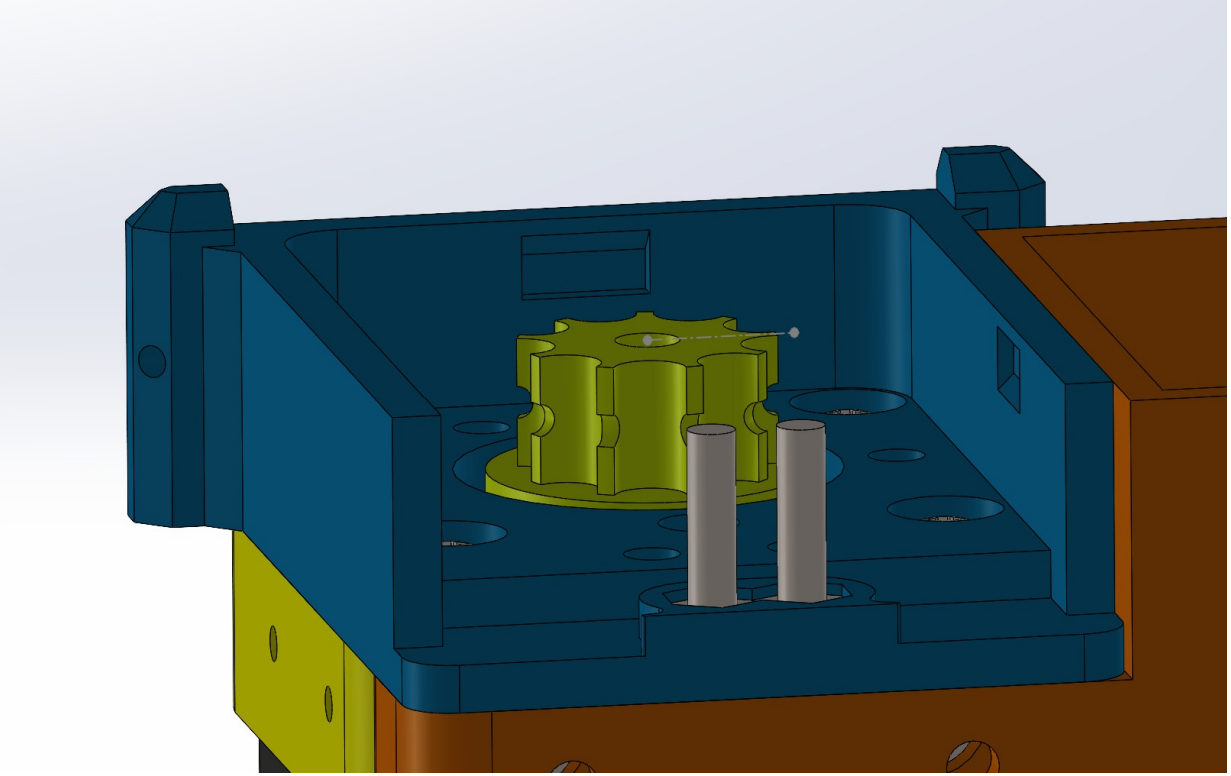


Figure 5a — gear position relative to Body

STEP 05 • PART 3 OF 3

Tighten the grub screw

Reach the grub screw (14) through the opening on the opposite side of Body (1).

Tighten the grub screw to secure gear9 (5) to the motor shaft.

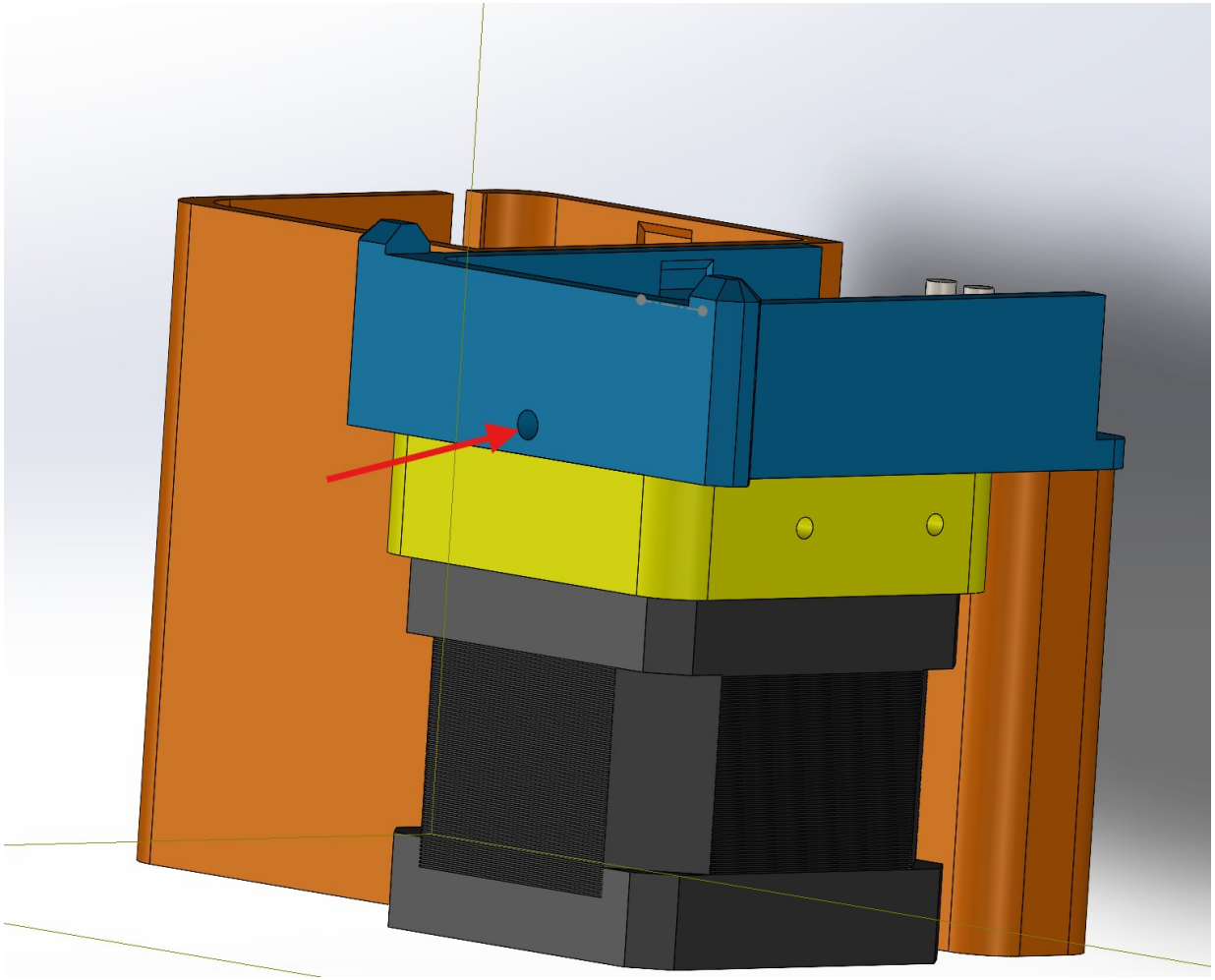


Figure 5b — access hole for tightening the grub screw

STEP 06 • PART 1 OF 2

Install Separator

Position Separator (7) in Body (1), as shown in the drawing.

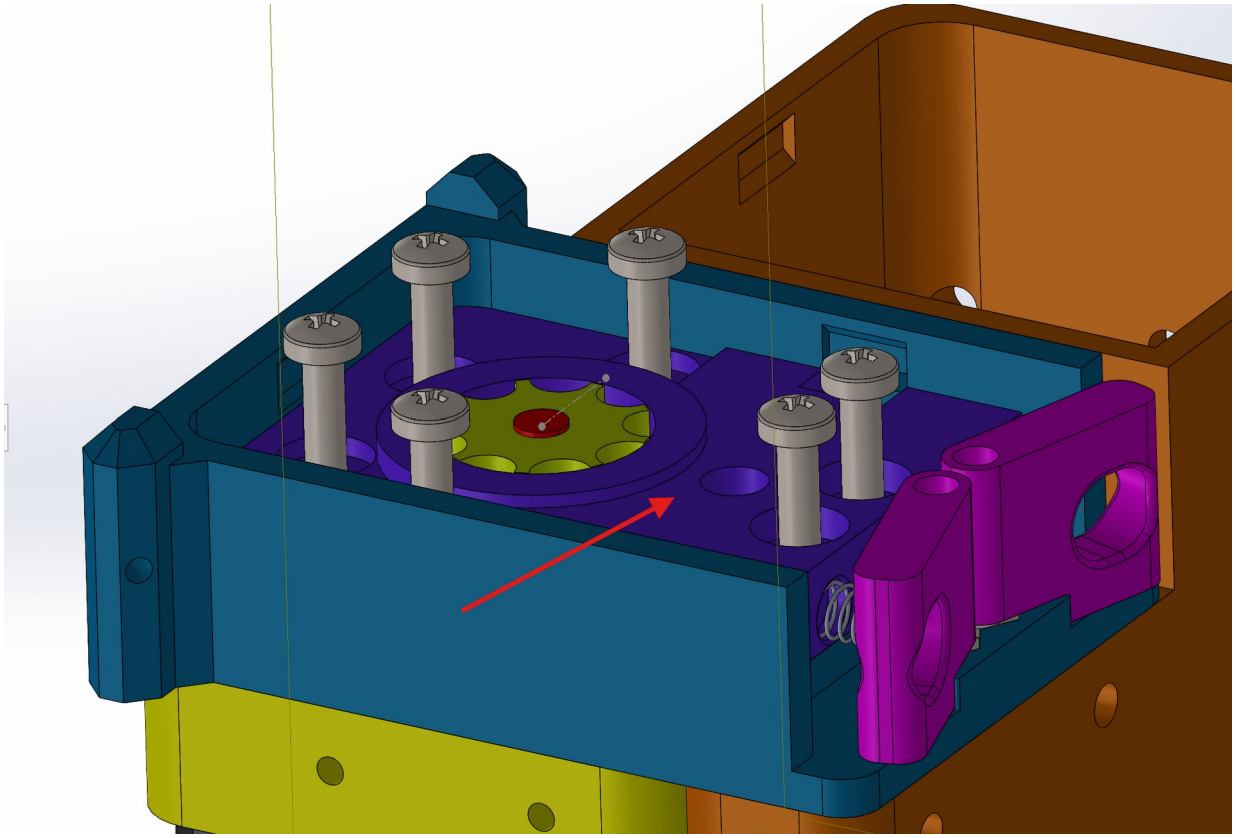


Figure 6 — install Separator

STEP 06 • PART 2 OF 2

Fasten Separator

Secure Separator (7) using the M3 × 10 screws (11), as shown.

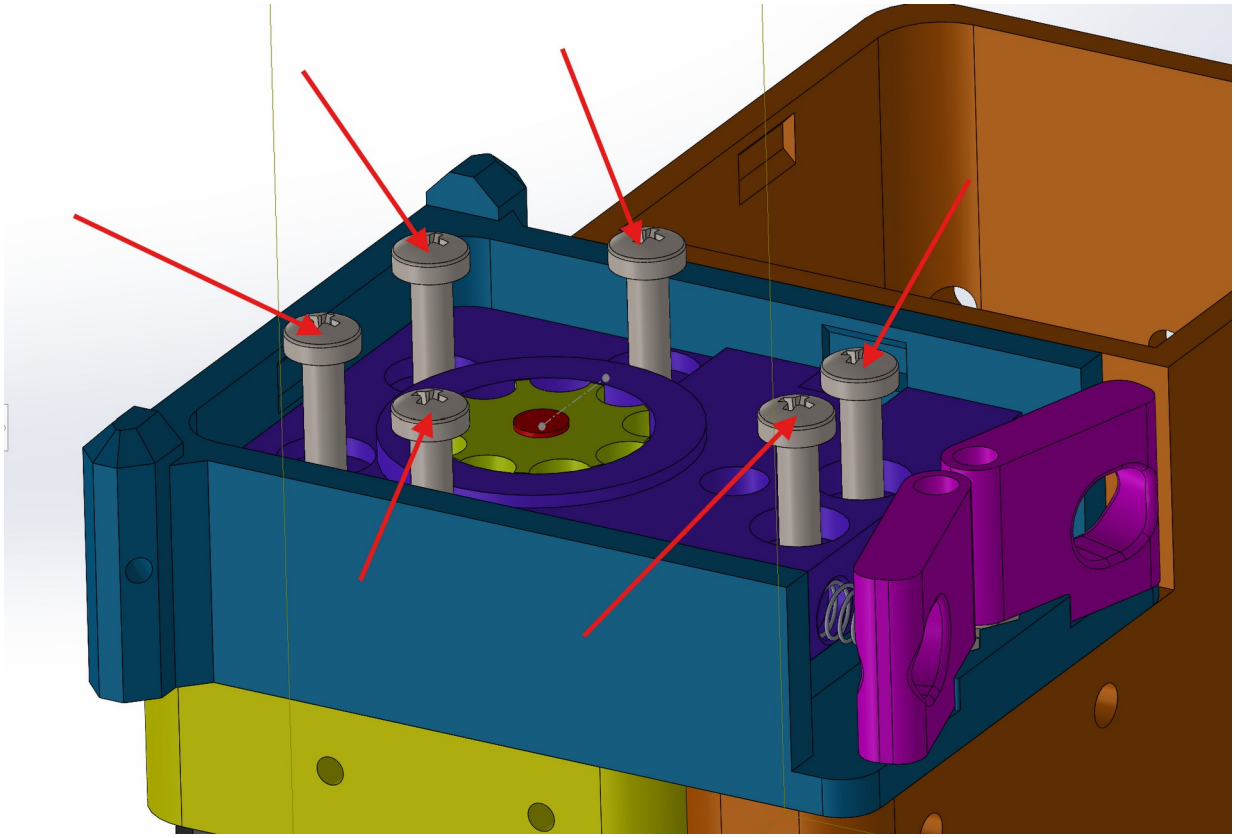


Figure 6a — locations of the M3 × 10 screws

STEP 07

Fit the springs and Gate

Insert the two springs (16) into their seats.

The original build used springs from a ballpoint pen. Depending on the spring type, you may need to shorten them.

Install Gate (6), following the orientation shown in the drawing.

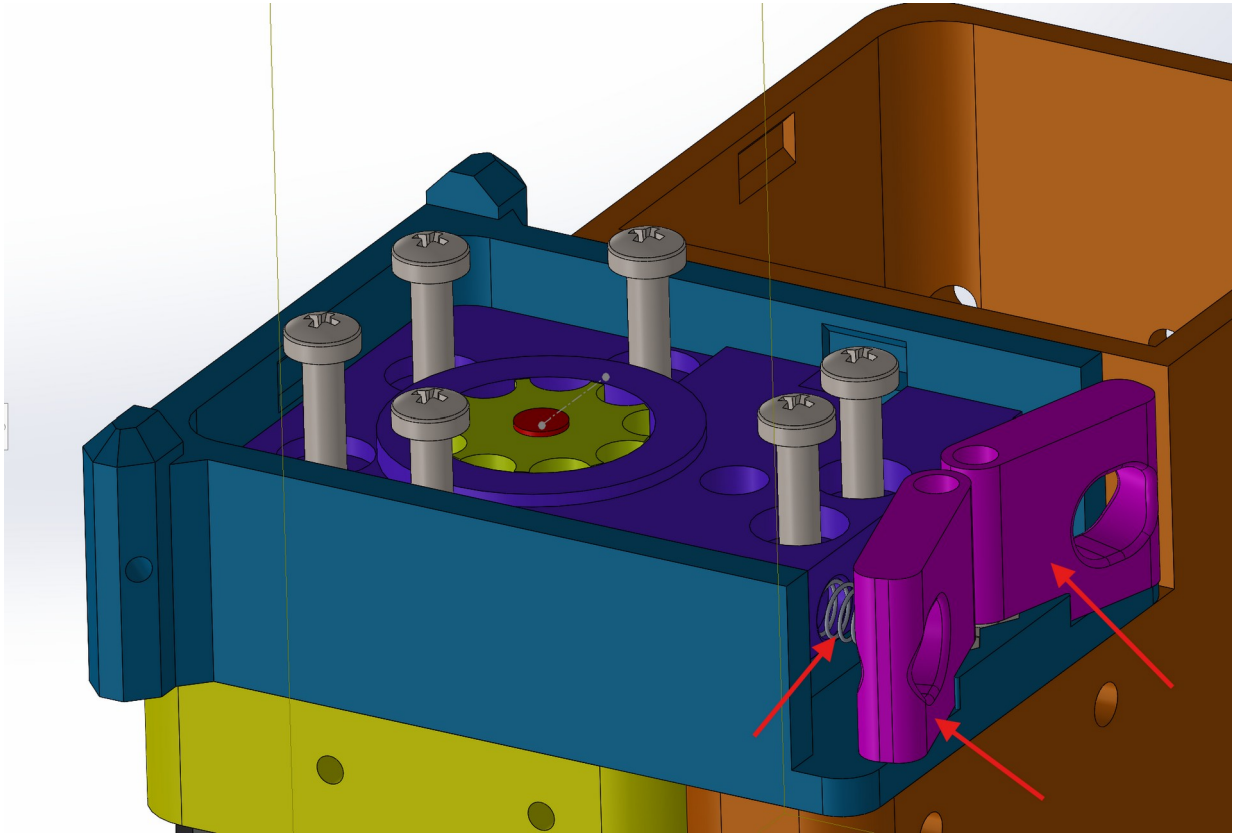


Figure 7 — spring locations and Gate

STEP 08

Install the AS5600 magnet

Place magnet (19), supplied with the AS5600 module (20), in the center of gear9 (5).

Adhesive may be needed to hold the magnet in place.

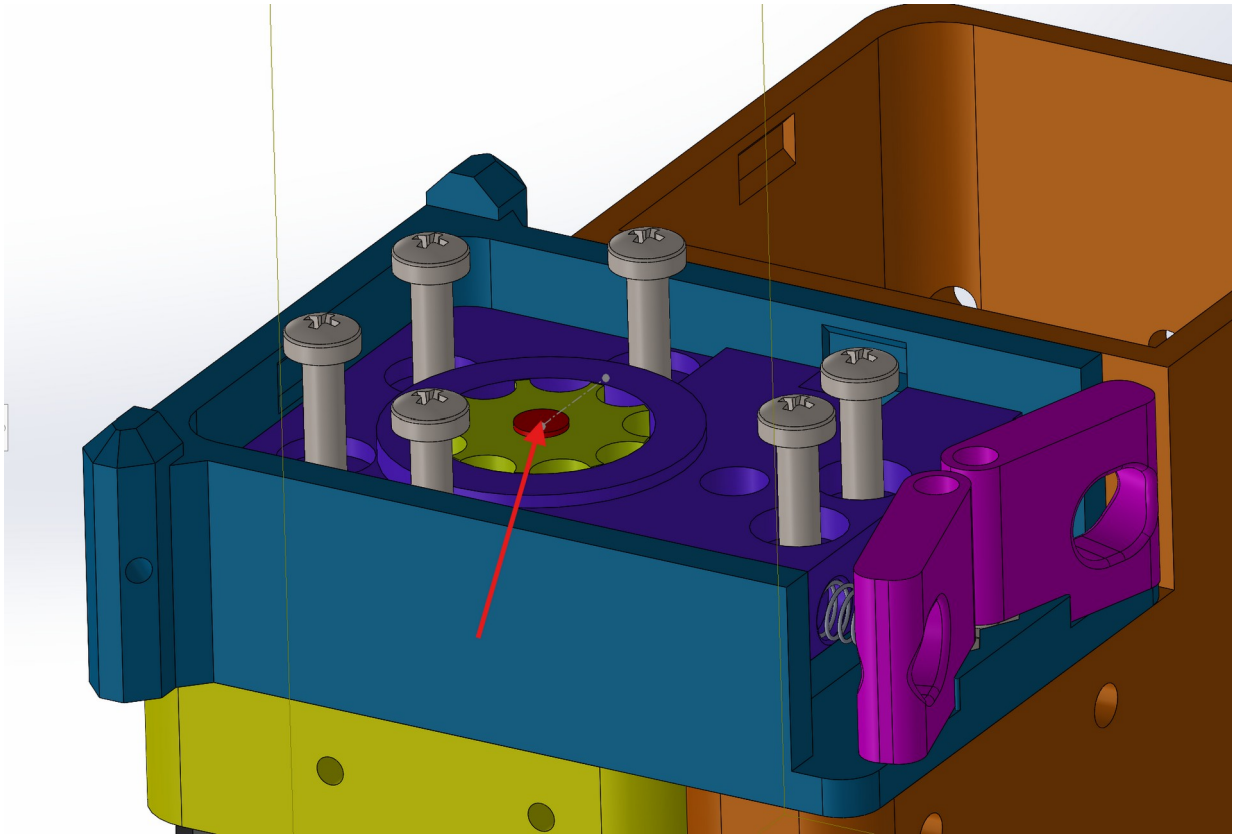


Figure 8 — magnet position at the gear center

STEP 09

Fit the sensor and end switches

Install AS5600 (20) and both End Switches (4) in CoverGearbox (2), as shown.

You may need to secure the sensor and switches with a small amount of adhesive.

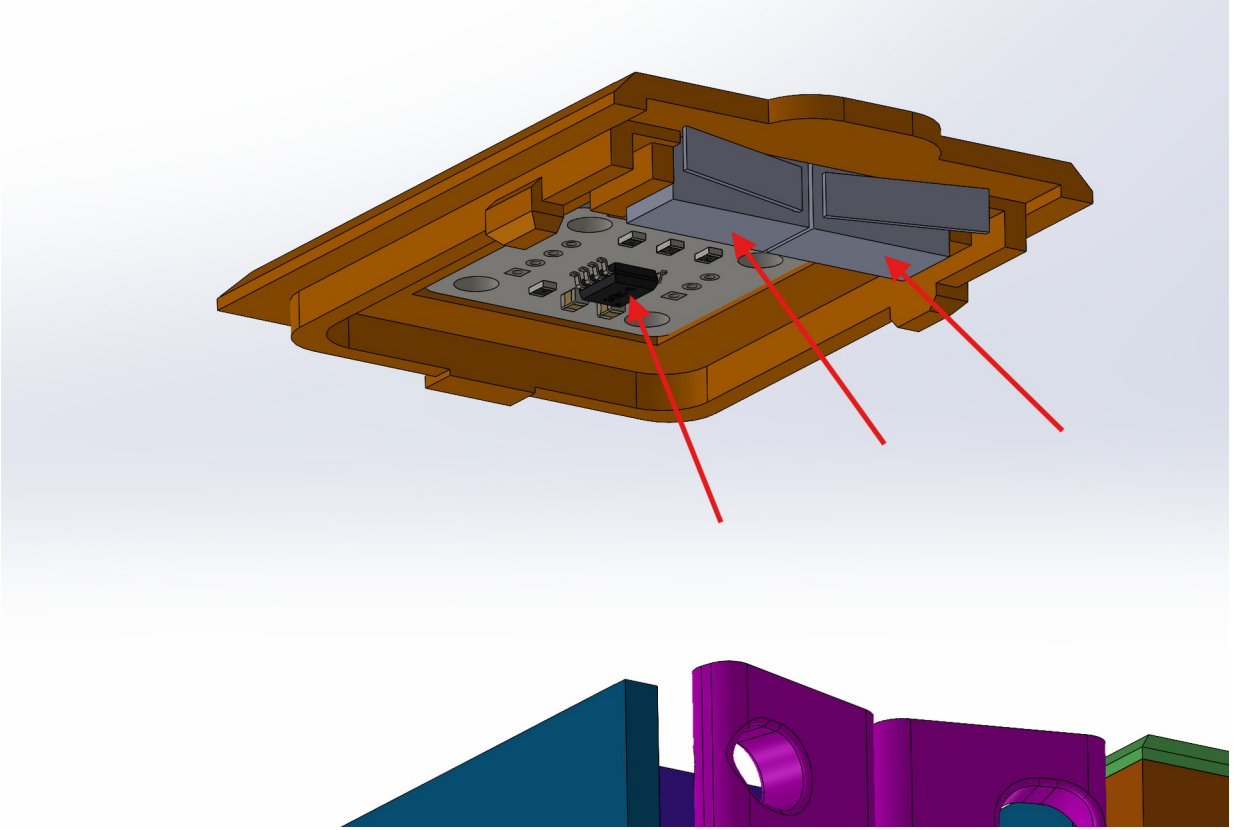


Figure 9 — sensor and switch positions inside CoverGearbox

STEP 10

Wire the sensor and switches

Solder the required wires: SDA, SCL, +, -, SW1 and SW2.

Connect the common terminals of the End Switches (4) to - (GND).

Use thin solid-core wire. Individual conductors from an Ethernet cable were used in the original build.

Bundle the wires running to the controller and feed them through the opening.

Make sure the wires cannot rub against gear9 (5). Then close CoverGearbox (2).

IMPORTANT

Keep all wires clear of the rotating gear.

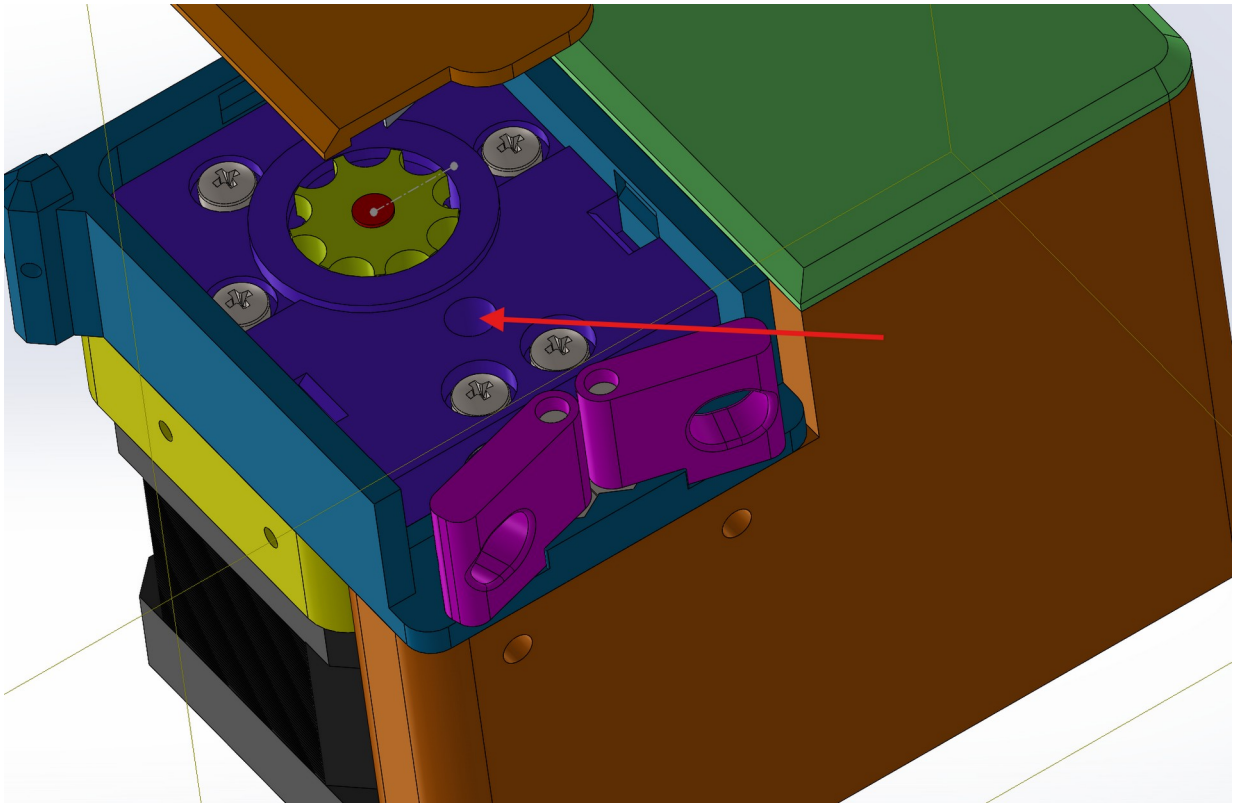


Figure 10 — wire routing through the opening

STEP 11

Install the controller and close the box

Flash the ESP32 board (21) with the required firmware before installing it.

Place it into electronicBox (17) with its USB connector facing down.

The enclosure is designed for the ESP32 header pins to be soldered to the top (component) side of the board.

Install the stepper motor driver, step-down converter and DC Jack (22). Connect everything according to the project pinout.

Close electronicBoxCover (24).

FIT WARNING

An ESP32 board with headers soldered on the underside may not fit the enclosure.

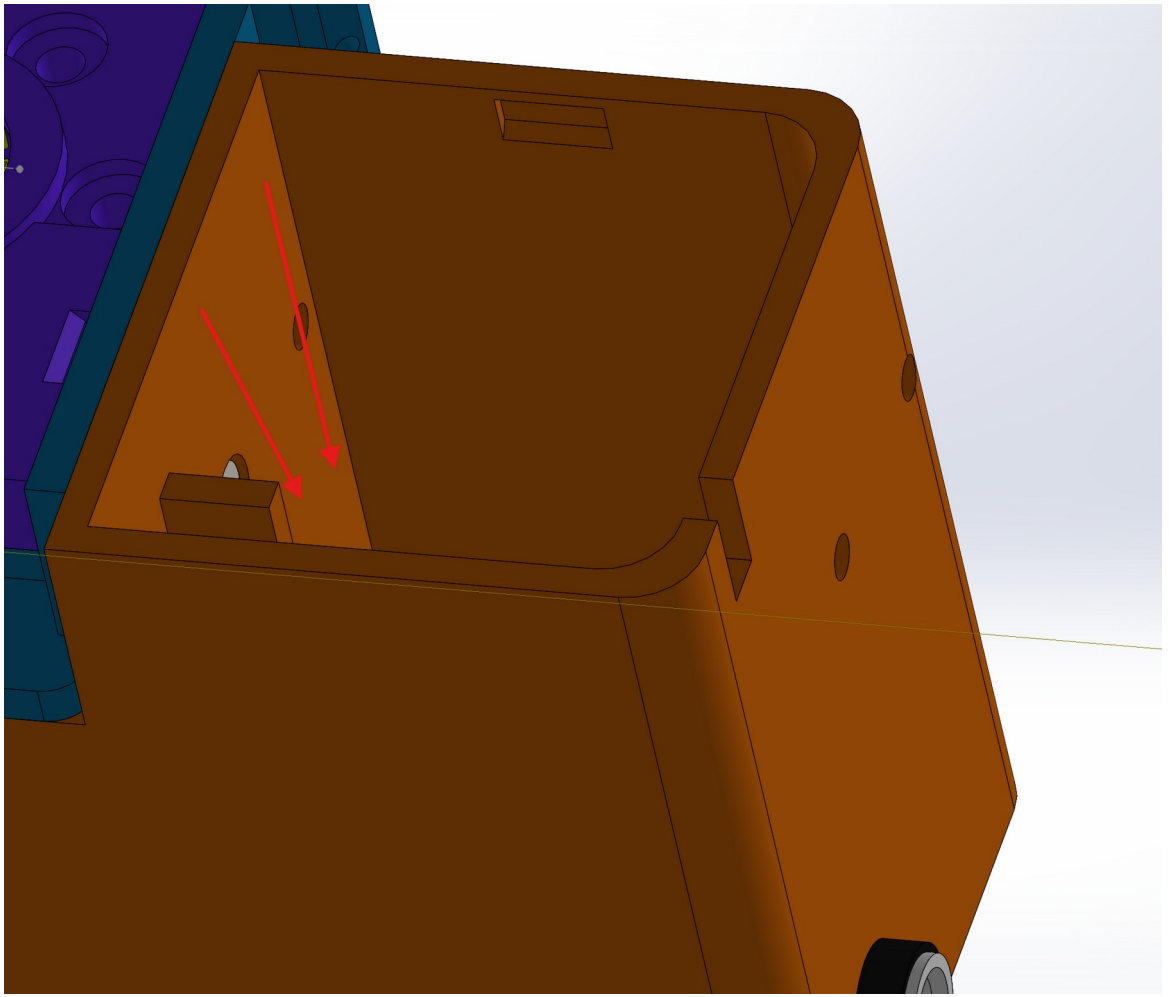


Figure 11 — ESP32 installation orientation